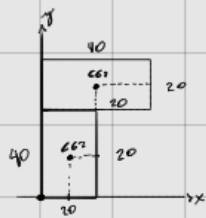


Momento Vetorial: força aplicada em um ponto em relação ao outro, para ter um momento

$$\vec{M}_O = \vec{r} \times \vec{F} \quad \vec{M}_O = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

* Para calcular Centro de gravidade:



Posição do Centro de gravidade

	A	$\bar{x}$	$A \cdot \bar{x}$	$\bar{y}$	$A \cdot \bar{y}$
1	800	20	16000	30	24000
2	900	40	36000	70	63000
Σ	1700	30	52000	40	87000

$$\bar{x}_{CG} = \frac{\Sigma A \bar{x}}{\Sigma A} = \frac{52000}{1700} = 30,59 \text{ m}$$

* Estes são os coordenados do Centro de gravidade

$$\bar{y}_{CG} = \frac{\Sigma A \bar{y}}{\Sigma A} = \frac{87000}{1700} = 51,18 \text{ m}$$



$\Delta \rightarrow$ Ponto x e y
O $\rightarrow$ Ponto oposto y

* Fixar A e B

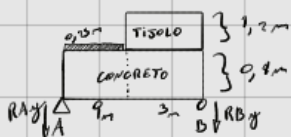
$$27000 \cdot 1 + 27000 \cdot 2 - 10000 \cdot 2 + R_{Ay} \cdot 3 = 0$$

$$R_{Ay} = 17866,66 \text{ N}$$

* Fixar A

$$10000 \cdot 1 - R_{By} \cdot 3 + 27000 \cdot 4 + 27000 \cdot 5,5 = 0$$

$$R_{By} = 108833,33 \text{ N}$$



$$V = (3,9 \cdot 2,0 \cdot 25) \text{ m}^3$$

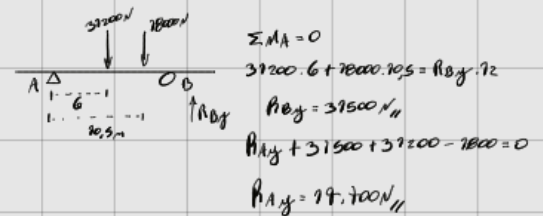
$$m = 2000 \cdot 9,9 = 19800 \text{ kg}$$

$$P_{total} = 198000 \text{ N}$$

$$V = 22,0 \cdot 9,9 \cdot 25 = 5490 \text{ m}^3$$

$$m = 2600 \text{ kg/m}^3 \cdot 5490 \text{ m}^3 = 14274000 \text{ kg}$$

$$P_{concreto} = 142740000 \text{ N}$$



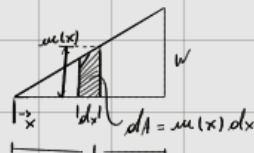
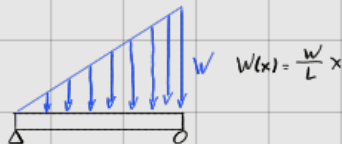
$$\Sigma M_A = 0$$

$$31500 \cdot 6 + 18000 \cdot 10,5 = R_{By} \cdot 12$$

$$R_{By} = 31500 \text{ N}$$

$$R_{Ay} + 31500 + 31500 - 18000 = 0$$

$$R_{Ay} = 14000 \text{ N}$$

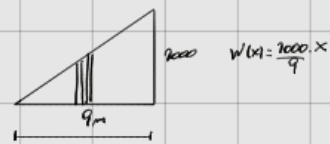


$$E_c = \text{AREA}$$

$$F_m = \int_0^L w(x) dx = \int_0^L \frac{W}{L} x dx = \frac{W}{L} \cdot \frac{x^2}{2} \Big|_0^L = \frac{W \cdot L}{2}$$

$$E_c = \frac{W \cdot L}{2}$$

$$\bar{x}_{CG} = \frac{2}{3} L$$



$$E_c = \int_0^9 w(x) dx = \int_0^9 \frac{2000}{9} x dx = \frac{2000}{9} \cdot \frac{x^2}{2} \Big|_0^9 = \frac{2000}{9} \cdot \frac{81}{2} = 90000 \text{ N}$$